

1 Time Evolution

We have seen the canonical pair $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$ and how the translation operator is built out of the momentum operator: $\hat{p}_i$ is said to be the generator of translations in the i -th spatial direction. For time evolution we have a similar picture, except that the “canonical pair” is time and energy. Time in QM is special as it is treated as an independent parameter, as opposed to an operator like $\hat{x}$ or $\hat{p}$. Still, the mathematical structure of time evolution will mimic that of spatial translations after we swap $\Delta x \rightarrow \Delta t$ and $\hat{p} \rightarrow \hat{H}$.

In particular, we want to build the time evolution operator as

$$\hat{U}(t, t_0) |\alpha, t_0\rangle = |\alpha, t\rangle \quad (1)$$

which I will sometimes write as

$$\hat{U}(t) |\alpha\rangle = |\alpha(t)\rangle \quad (2)$$

with the understanding that $t_0 = 0$ is some initial time. The distinction should be clear from the context.

Some properties we would like $\hat{U}(t, t_0)$ include:

$$\hat{U}(t_0, t_0) = \hat{I} \quad (3)$$

$$\hat{U}(t_2, t_1) \hat{U}(t_1, t_0) = \hat{U}(t_2, t_0) \quad (4)$$

$$\hat{U}(t_1, t_0)^\dagger = \hat{U}(t_0, t_1) \quad \text{or} \quad \hat{U}(t, t_0)^\dagger \hat{U}(t, t_0) = \hat{I}. \quad (5)$$

Note especially the unitarity condition: as you proved in HW4, this is essential to maintain the system’s total probability over time (“probability is conserved”).

In analogy to the translation operator, we write

$$\hat{U}(t_0 + dt, t_0) = \hat{I} - \frac{i}{\hbar} \hat{H}(t_0) dt, \quad (6)$$

where $\hbar$ here is the same constant that appeared in the translation operator (recall, we had to identify $\hat{K} = \hat{p}/\hbar$ in that context).

From the above, it follows that

$$\hat{U}(t + dt, t) \hat{U}(t, t_0) = \left(\hat{I} - \frac{i}{\hbar} \hat{H}(t) dt \right) \hat{U}(t, t_0) \quad (7)$$

and rearranging,

$$\hat{U}(t + dt, t_0) - \hat{U}(t, t_0) = -\frac{i}{\hbar} \hat{H}(t) dt \hat{U}(t, t_0). \quad (8)$$

Without apologising to mathematicians, we divide both sides by dt , interpret the different in the left-hand side as $d\hat{U}(t, t_0)/dt$, and then take the limit $dt \rightarrow 0$. We get the **time-dependent Schrödinger equation (TDSE) for the time-evolution operator**:

$$\boxed{i\hbar \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H}(t) \hat{U}(t, t_0), \quad \hat{U}(t_0, t_0) = \hat{I}.} \quad (9)$$

If you know how to solve this equation, you have the full time evolution of the quantum system. Give me your initial conditions, $|\alpha(t_0)\rangle$, and you can get the state at any later time:

$$|\alpha(t)\rangle = \hat{U}(t, t_0) |\alpha(t_0)\rangle. \quad (10)$$

We would also like to have an equation for the state vector itself. To that end, multiply both sides of Eq. (7) by $|\alpha(t_0)\rangle$ from the right, rearrange, and we get:

$$|\alpha(t + dt)\rangle - |\alpha(t)\rangle = -\frac{i}{\hbar} \hat{H}(t) dt |\alpha(t)\rangle, \quad (11)$$

which through the same mathematical sins, gives us the **time-dependent Schrödinger equation (TDSE) for the state vector**:

$$\boxed{i\hbar \frac{\partial}{\partial t} |\alpha(t)\rangle = \hat{H}(t) |\alpha(t)\rangle.} \quad (12)$$

In practice, these two equations are equivalent.

Now, let us go back to the form of the evolution operator in Eq. (6). For a finite time step, we repeat our logic from the translation discussion:

$$\hat{U}(t_0 + \Delta t, t_0) |\alpha(t_0)\rangle = \left(\mathbb{1} - \frac{i}{\hbar} \hat{H}(t_0) \frac{\Delta t}{N} \right)^N |\alpha(t_0)\rangle \quad (13)$$

$$= \left(\mathbb{1} - \frac{i}{\hbar} \hat{H}(t_0) \frac{\Delta t}{N} N - \frac{\hat{H}^2(t_0)}{\hbar^2} \frac{(\Delta t)^2}{N^2} N(N-1)/2 + \dots \right) |\alpha(t_0)\rangle \quad (14)$$

The Hamiltonian is evaluated at t_0 . Note the second line is just multiplying the binomial expansion of the first line. The N factors in the numerator are just the combinatorial coefficients from the binomial expansion: $\binom{N}{k} = \frac{N!}{k!(N-k)!}$ for the k -th term. Taking N to infinity, we get the familiar exponential form:

$$\hat{U}(t_0 + \Delta t, t_0) = \left(\mathbb{1} - \frac{i}{\hbar} \hat{H}(t_0) \Delta t + \frac{1}{2} \left(\frac{i}{\hbar} \hat{H}(t_0) \Delta t \right)^2 + \dots \right) = e^{-\frac{i}{\hbar} \hat{H}(t_0) \Delta t}. \quad (15)$$

(I am doing this as a warm up for what comes next).



There is an important assumption in Eq. (15): the Hamiltonian is assumed to be constant during the time interval Δt . In reality, our daisy chain of infinitesimal time evolution operators would look more like

$$\hat{U}(t_0 + \Delta t, t_0) \approx \left(\mathbb{1} - \frac{i}{\hbar} \hat{H} \left(t_0 + \Delta t - \frac{\Delta t}{N} \right) \frac{\Delta t}{N} \right) \dots \left(\mathbb{1} - \frac{i}{\hbar} \hat{H}(t_0) \frac{\Delta t}{N} \right) \neq e^{-\frac{i}{\hbar} \hat{H}(t_0) \Delta t}. \quad (16)$$

Now, two things can happen: 1) the Hamiltonian can be different at different times, in which case we cannot exponentiate as easily as in Eq. (15) (the cure here will be straightforward, we will only need an integral in the exponent), but also 2) the Hamiltonian at different times may not commute with each other(!), in which case the combinatorics at any given order in $\hat{H}$ become much more cumbersome (the cure here is more subtle and involves the time-ordering operator that we discuss below). Alas, such is time in QM: respecting causality will require little more book-keeping.

2 Time-Independent Hamiltonians

If the time evolution of a closed system is governed by the time-dependent Schrödinger equation

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (17)$$

For a time-independent Hamiltonian $\hat{H}(t) = \hat{H}$, we can also find the time-evolved state directly (treat this as a differential equation with constant coefficients, as opposed to finding $\hat{U}(t)$ and then evolving it):

$$|\psi(t)\rangle = e^{-\frac{i}{\hbar} \hat{H} t} |\psi(0)\rangle, \quad (18)$$

A similar approach gives using Eq. (9)

$$\hat{U}(t) = e^{-\frac{i}{\hbar} \hat{H} t}. \quad (19)$$

First, we ask: what is the eigenbasis of $\hat{U}(t)$? This should be a good basis to time evolve our system as the exponential of the operator becomes a single time-dependent number (the eigenvalue of $\hat{U}(t)$). To find this basis, note that

$$[\hat{H}, \hat{U}(t)] = 0 \quad (20)$$

so $\hat{H}$ and $\hat{U}(t)$ share the same eigenbasis (as per your HW1). So the eigenbasis of $\hat{U}(t)$ is the same as the energy eigenbasis of $\hat{H}$:

$$\hat{H} |E_n\rangle = E_n |E_n\rangle, \quad (21)$$

such that

$$\hat{U}(t) |E_n\rangle = \left(\mathbb{1} - i \frac{\hat{H}t}{\hbar} + \dots \right) |E_n\rangle = \left(\mathbb{1} - i \frac{E_n t}{\hbar} + \dots \right) |E_n\rangle = e^{-\frac{i}{\hbar} E_n t} |E_n\rangle. \quad (22)$$

What are the matrix elements of the time evolution operator in the energy eigenbasis? Inserting a completeness relation in the energy eigenbasis, we get

$$\hat{U}(t) = \sum_{m,n} |E_n\rangle \langle E_n| \hat{U}(t) |E_m\rangle \langle E_m| = \sum_{n,m} |E_n\rangle \langle E_n| e^{-\frac{i}{\hbar} E_m t} |E_m\rangle \langle E_m| = \sum_n e^{-\frac{i}{\hbar} E_n t} |E_n\rangle \langle E_n|. \quad (23)$$

where again we used the eigenvalue equation that $\hat{U}(t) |E_m\rangle = e^{-\frac{i}{\hbar} E_m t} |E_m\rangle$. In other words: the time evolution operator is diagonal in the energy basis (it is its own eigenbasis), and its matrix elements are complex phases (it must be since this is a unitary operator).

We can now evolve any state by expanding it in the energy eigenbasis:

$$|\psi(0)\rangle = \sum_n c_n |E_n\rangle, \quad c_n = \langle E_n | \psi(0) \rangle, \quad (24)$$

so that the time-evolved state is

$$|\psi(t)\rangle = \hat{U}(t) |\psi(0)\rangle = \sum_n c_n e^{-\frac{i}{\hbar} E_n t} |E_n\rangle. \quad (25)$$

We may also want to define $c_n(t) = \langle E_n | \psi(t) \rangle = c_n e^{-\frac{i}{\hbar} E_n t}$, the time-dependent expansion coefficients in the energy eigenbasis.



Now we are ready to write the TDSE in a given basis. Note that both Eq. (9) and Eq. (12) are basis independent. As you saw in HW1, they give rise to the usual equation in position, momentum, energy, etc, after projecting onto the corresponding basis states. All we need is Eq. (12) and the definition of the energy-basis wavefunction

$$c_n(t) = \langle E_n | \psi(t) \rangle. \quad (26)$$

Take the inner product of $|E_n\rangle$ with both sides of the TDSE, we get

$$\boxed{i\hbar \frac{d}{dt} c_n(t) = E_n c_n(t)}. \quad (27)$$

You should know how to derive this. This is a very simple equation, but it assumes you have already solved the eigenvalue problem for $\hat{H}$, i.e., you know the energy eigenstates $|E_n\rangle$, their corresponding eigenvalues E_n , and how to write your initial state $|\psi(0)\rangle$ in terms of these eigenstates. In practice, this is not always doable.

2.1 Stationary States

Take an operator that is compatible with the Hamiltonian, $[\hat{H}, \hat{A}] = 0$. We want to show that its eigenvalues are conserved: if our system is initially in an eigenstate of $\hat{A}$ with eigenvalue a , it remains an eigenstate of $\hat{A}$ with the same eigenvalue a for all times. This is obviously the case for the energy basis, so the energy eigenstates are stationary states.

To see that, take

$$\hat{A} |a\rangle = a |a\rangle, \quad (28)$$

so that

$$\hat{A} |a(t)\rangle = \hat{A} \hat{U}(t) |a\rangle = \hat{U}(t) \hat{A} |a\rangle = a \hat{U}(t) |a\rangle = a e^{-\frac{i}{\hbar} \hat{H} t} |a\rangle = a |a(t)\rangle. \quad (29)$$

where $[\hat{U}(t), \hat{A}] = 0$ follows from $[\hat{H}, \hat{A}] = 0$. The expectation value of $\hat{A}$ in the state $|a(t)\rangle$ is unchanged over time:

$$\langle \hat{A} \rangle_{a(t)} = \langle a(t) | \hat{A} | a(t) \rangle = \langle a | \hat{A} | a \rangle = a. \quad (30)$$

Note that I am writing the time dependence in $\langle \hat{A} \rangle_{a(t)}$ to indicate that it is the expectation value of $\hat{A}$ in the **time-evolved** state $|a(t)\rangle$. If a is non-degenerate, there is only one such state, so $|a(t)\rangle$ can only differ from $|a\rangle$ by a phase: the state is stationary, and $|a\rangle$ must in fact be an energy eigenstate (HW1). If a is degenerate, the state can move around inside the a -eigenspace. We will see important examples of degenerate states when solving the hydrogen atom later in the course.

Contrast this with the general case for some other generic operator $\hat{B}$ that may or may not be compatible with the Hamiltonian:

$$\langle \hat{B} \rangle_{b(t)} = \langle b(t) | \hat{B} | b(t) \rangle = \langle b | \hat{U}^\dagger(t) \hat{B} \hat{U}(t) | b \rangle = \sum_{i,j} \langle b | E_i \rangle \langle E_i | \hat{B} | E_j \rangle \langle E_j | b \rangle e^{\frac{i}{\hbar} (E_i - E_j) t}. \quad (31)$$

Using the notation $c_i = \langle E_i | b \rangle$, we can rewrite the expectation value as

$$\boxed{\langle \hat{B} \rangle_{b(t)} = \sum_{i,j} c_i^* c_j \langle E_i | \hat{B} | E_j \rangle e^{\frac{i}{\hbar} (E_i - E_j) t}.} \quad (32)$$

This is quite general, but not necessarily useful if we don't already know c_i and the matrix elements $\langle E_i | \hat{B} | E_j \rangle$. Note that if $\langle E_i | \hat{B} | E_j \rangle = b_i \delta_{ij}$, i.e., if $\hat{B}$ is diagonal in the energy eigenbasis, the expectation value simplifies to

$$\langle \hat{B} \rangle_{b(t)} = \sum_i |c_i|^2 b_i, \quad (33)$$

which is time-independent. But this is equivalent to saying that the operator $\hat{B}$ commutes with the Hamiltonian, $[\hat{B}, \hat{H}] = 0$ since they are diagonalized by same eigenbasis (for non-degenerate energy levels).



Go back to our notes on spin precession and the time-dependent Bloch vector to convince yourself that they are using the machinery above. This is S&N Sec. 2.1.5.

2.2 Correlation Amplitude and Transition Probability

The correlation amplitude between an initial state $|a\rangle$ and a state $|b\rangle$ at time t is defined as

$$C_{ab}(t) = \langle b | a(t) \rangle = \langle b | \hat{U}(t) | a \rangle. \quad (34)$$

Squaring its modulus gives the transition probability between the two states (same maths behind your HW 5):

$$P_{ab}(t) = |C_{ab}(t)|^2 = |\langle b | \hat{U}(t) | a \rangle|^2. \quad (35)$$

Now, the key observation about stationary states: if we start out in a state $|E_i\rangle$ which is an eigenstate of the Hamiltonian, the time evolution is just a phase, so the correlation amplitude is always diagonal:

$$C_{ij}(t) = \langle E_j | E_i(t) \rangle = \langle E_j | \hat{U}(t) | E_i \rangle = \langle E_j | e^{-\frac{i}{\hbar} \hat{H} t} | E_i \rangle = e^{-\frac{i}{\hbar} E_i t} \delta_{ij}, \quad \implies P_{ij}(t) = \delta_{ij}. \quad (36)$$

That is: the state never evolves to be in any other energy eigenstate ever. This is unrealistic because in the real world we 1) usually cannot produce energy eigenstates exactly or 2) the Hamiltonian is almost never exactly time independent.



The general form of the correlation amplitude for a generic not necessarily stationary state is

$$C_{ab}(t) = \langle b | a(t) \rangle = \langle b | \hat{U}(t) | a \rangle = \sum_i \langle b | E_i \rangle \langle E_i | a \rangle e^{-\frac{i}{\hbar} E_i t} = \sum_i c_i d_i^* e^{-\frac{i}{\hbar} E_i t}, \quad (37)$$

where $c_i = \langle E_i | a \rangle$ and $d_i = \langle E_i | b \rangle$.

Following S&N, let us consider a continuous system. We could proceed with the following continuous expansion into the energy basis,

$$|a\rangle = \int dE |E\rangle \langle E | a \rangle = \int dE c(E) |E\rangle, \quad (38)$$

and then write

$$C_{ab}(t) = \langle b | a(t) \rangle = \langle b | \hat{U}(t) | a \rangle = \int dE \langle b | E \rangle \langle E | a \rangle e^{-\frac{i}{\hbar} E t} = \int dE c(E) d^*(E) e^{-\frac{i}{\hbar} E t}, \quad (39)$$

which for $a = b$ (how the state overlaps with itself over time) gives the survival amplitude:

$$C_{aa}(t) = \int dE |c(E)|^2 e^{-\frac{i}{\hbar} E t}. \quad (40)$$

This is almost S&N Eq. (2.70), except S&N write it with an extra density of states $\rho(E)$,

$$C_{aa}(t) = \int dE \rho(E) |g(E)|^2 e^{-\frac{i}{\hbar} E t}, \quad (41)$$

where $g(E)$ is not the same as our $c(E)$: the two differ by how the energy states are normalized. S&N start from discrete, Kronecker delta normalization $\langle E_i | E_j \rangle = \delta_{ij}$ (think of a particle in a very large box with discrete values of momenta, as we will see later) with $c_j = \langle E_j | a \rangle$, and turn the sum into an integral by counting how many levels fall in $[E, E + dE]$:

$$\sum_j \longrightarrow \int dE \rho(E), \quad |c_j|^2 \longrightarrow |g(E)|^2. \quad (42)$$

Our $|E\rangle$ are instead normalized as $\langle E | E' \rangle = \delta(E - E')$, which absorbs the level density into the normalization of the kets, $|c(E)|^2 = \rho(E) |g(E)|^2$. Either way $C_{aa}(0) = \int dE |c(E)|^2 = \int dE \rho(E) |g(E)|^2 = 1$.

Note that a function like ρ is not needed for position or momentum because those states come with a flat measure $d^3\mathbf{x}$ and $d^3\mathbf{p}$. The density of states appears when we relabel states by their energy: it contains the Jacobian of the change of variables and potential degeneracies. For example, for a free particle in 3D every momentum on the sphere $|\mathbf{p}| = \sqrt{2mE}$ has the same energy.



Read S&N 2.1.4 - 2.1.7. and 2.2.2

3 Heisenberg Picture

When we calculated the expectation value of an operator $\hat{A}$, we wrote

$$\langle \hat{A} \rangle_{\psi(t)} = \langle \psi(t) | \hat{A} | \psi(t) \rangle = \langle \psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \psi(0) \rangle. \quad (43)$$

Writing things in this way gives us another method to interpret the expectation value: instead of evolving the states, we could imagine evolving the operators with $\hat{A}(t) = \hat{U}^\dagger(t) \hat{A} \hat{U}(t)$ while keeping the states fixed. In that case, the expectation value could be something like

$$\langle \hat{A}(t) \rangle_{\psi(0)} = \langle \psi(0) | \hat{A}(t) | \psi(0) \rangle = \langle \psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \psi(0) \rangle. \quad (44)$$

The same thing as Eq. (43). Because all we have access to in QM are expectation values, the distinction between this time-dependent operator and the time-dependent state is irrelevant in practice.

The picture of time-evolved operators is called the **Heisenberg picture**:

$$\hat{A}_H(t) = \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t), \quad (45)$$

where we added the subscript H to indicate the Heisenberg picture and the subscript S to indicate the Schrödinger picture operator. The states themselves do not evolve in the Heisenberg picture; they remain as $|\psi(0)\rangle$. In S&N, they go further and write the subscripts for the states as well:

$$|\psi_H(t)\rangle = \hat{U}^\dagger(t) |\psi_S(t)\rangle = \hat{U}^\dagger(t) \hat{U}(t) |\psi(0)\rangle = |\psi(0)\rangle. \quad (46)$$

This is a bit redundant, but it gives a useful perspective: the Heisenberg picture states is obtained by “winding back” the time-dependent states to their initial configuration.

3.1 Heisenberg Equation of Motion

We know the TDSE for the time evolution operator Eq. (9) and of a state Eq. (12). We can use that to find a type of TDSE for any Heisenberg picture operator. To get that, differentiate the Heisenberg picture operator with respect to time:

$$\frac{d}{dt} \hat{A}_H(t) = \frac{d}{dt} (\hat{U}^\dagger(t) \hat{A}_S \hat{U}(t)) = \frac{d\hat{U}^\dagger(t)}{dt} \hat{A}_S \hat{U}(t) + \hat{U}^\dagger(t) \hat{A}_S \frac{d\hat{U}(t)}{dt} + \hat{U}^\dagger(t) \frac{\partial \hat{A}_S}{\partial t} \hat{U}(t) \quad (47)$$

Using Eq. (9) and its Hermitian conjugate, we get

$$\begin{aligned} \frac{d}{dt} \hat{A}_H(t) &= \frac{1}{i\hbar} (-\hat{U}^\dagger(t) \hat{H} \hat{A}_S \hat{U}(t) + \hat{U}^\dagger(t) \hat{A}_S \hat{H} \hat{U}(t)) + \hat{U}^\dagger(t) \frac{\partial \hat{A}_S}{\partial t} \hat{U}(t) \\ &= \frac{1}{i\hbar} [\hat{A}_H(t), \hat{H}_H(t)] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H \end{aligned} \quad (48)$$

where $\hat{H}_H(t) = \hat{U}^\dagger(t) \hat{H} \hat{U}(t)$. For time-independent Hamiltonians, $\hat{H}_H(t) = \hat{H}$ since $[\hat{H}, \hat{U}(t)] = 0$. So, we arrive at the **Heisenberg equation of motion** for time-independent Hamiltonians:

$$\boxed{\frac{d}{dt} \hat{A}_H(t) = \frac{1}{i\hbar} [\hat{A}_H(t), \hat{H}] + \left(\frac{\partial \hat{A}_S}{\partial t} \right)_H}. \quad (49)$$

Compared to S&N we kept the possibility that $\hat{A}_S$ has explicit time dependence. Even though it is in the Schrödinger picture, it can still have explicit time dependence.



Perhaps the most important lesson from Eq. (49) is that operators that commute with the Hamiltonian (and which have no explicit time dependence) are constants of motion in the Heisenberg picture:

$$[\hat{A}_H(t), \hat{H}] = 0 \quad \Rightarrow \quad \frac{d}{dt} \hat{A}_H(t) = 0.$$

We already knew that from our discussion about stationary states: operators that are compatible with the Hamiltonian can be diagonalized with the energy eigenstates, and therefore their expectation values in these states do not change with time as per Eq. (30).

Free particle discussion
Boost generators



Read Sections 2.2.3 to 2.2.5.